

IRIS FPU data sheet and design overview

1 Overview

The IRIS FPU is an IEEE-754 floating point unit implementing all floating point operations as required by the SPARC V8 standard. It supports both single- and double-precision operations including de-normalized numbers. All four IEEE-754 rounding modes are supported. The interface to the FPU has been modeled after the Meiko FPU from the Sun Microsparc processor. The IRIS FPU is iterative, only one instruction is calculated at a time.

2 Instructions

The IRIS FPU implements all necessary FPU instructions required by the SPARC V8 standard.

Instruction	Opcode	Op1	Op2	Result	Exceptions	Description
FADDS	041	SP	SP	SP	NV, OF, UF, NX	Add single
FADDD	042	DP	DP	DP	NV, OF, UF, NX	Add double
FSUBS	045	SP	SP	SP	NV, OF, UF, NX	Subtract single
FSUBD	046	DP	DP	DP	NV, OF, UF, NX	Subtract double
FMULS	049	SP	SP	SP	NV, OF, UF, NX	Multiply single
FMULD	04A	DP	DP	DP	NV, OF, UF, NX	Multiply double
FSMULD	069	SP	SP	DP	NV	Multiply single to double
FDIVS	04D	SP	SP	SP	NV, OF, DZ, UF, NX	Divide single
FDIVD	04E	DP	DP	DP	NV, OF, DZ, UF, NX	Divide double
FSQRTS	029	-	SP	SP	NV, NX	Square root single
FSQRD	02A	-	DP	DP	NV, NX	Square root double
FITOS	0C4	-	INT32	SP	NX	Convert 32-bit integer to single
FITOD	0C8	-	INT32	DP	-	Convert 32-bit integer to double
FSTOI	0D1	-	SP	INT32	NV, NX	Convert single to 32-bit integer
FDTOI	0D2	-	DP	INT32	NV, NX	Convert double to 32-bit integer
FSTOD	0C9	-	SP	DP	NV	Convert single to double
FDTOS	0C6	-			NV, OF, UF, NX	Convert double to single
FCMPS	251	SP	SP	CC	NV	Compare single
FCMPD	252	DP	DP	CC	NV	Compare double
FCMPES	255	SP	SP	CC	NV	Compare single and exception if unordered
FCMPED	256	DP	DP	CC	NV	Compare double and exception if unordered
FABSS	009	-	SP	SP	-	Absolute value single
FNEGS	005	-	SP	SP	-	Negate single
FMOVS	001	-	SP	SP	-	Move (copy) single

3 Signals

The table below list the IRIS FPU signals. All signals are active high. All inputs are sampled on the rising edge of CLOCK, while all outputs are driven on the rising edge of CLOCK.

Signal	I/O	Description
CLOCK	I	Clock
FPINST[9:0]	I	FPU instruction opcode
FPOP	I	Load FPU opcode
FPLD	I	Load FPU operands and start execution
RESET	I	Synchronous reset, active high
FPRF_DOUT1[63:0]	I	Operand 1
FPRF_DOUT2[63:0]	I	Operand 2
ROUNDINGMODE	I	Rounding mode. 00 – round to nearest, 01 – round to zero, 10 – round to +inf, 11 – round to -inf
FPBUSY	O	Indicates the the FPU is executing
FRACRESULT[54:3]	O	Fraction result
EXPRESULT[10:0]	O	Exponent result
SIGNRESULT	O	Sign result
SNNOTDB	O	Indicates that results is single precision or 32-bit integer
EXCEP[5:0]	O	FPU exceptions
CONDITIONCODES[1:0]	O	Result (condition codes) of a compare instruction 00 – equal 01 – operand1 < operand2 10 – operand1 > operand2 11 - unordered

Single-precision operands should be placed in FPRF_DOUT1/2[63:32]. If the result is single-precision or 32-bit integer (SNNOTDB = 1), the value is given by:

Result = SIGNRESULT & EXPRESULT[7:0] & FRACRESULT[54:32].

If the result is double-precision (SNNOTDB=0), the value is given by:

Result = SIGNRESULT & EXPRESULT[10:0] & FRACRESULT[54:3].

If an instruction only takes one operand, it should be placed on FPRF_DOUT2.

4 Generics

The operation of the IRIS FPU can be configured through five generics.

Generic	Default value	Description
Fpga	0	If set to 1, the sticky bit generation is implemented with an adder. This can improve timing on FPGA technologies but should not be used on ASICs.
Busydelay	0	The delay in clock cycles from FpBusy to valid result data. Best timing is achieved with setting this generic to 0, which indicates that FpBusy is driven low in the same cycle as the result data is valid. If set to 1, FpBusy is driven low one clock before the result data is valid. Should only be set to 1 if direct compatibility with the Meiko is required.
Srtopt	3	Indicates how many bits the SRT divider can evaluate per cycle. A value of 3 provides highest throughput, while lower numbers (0 – 2) will reduce area and throughput of the divider.
TinyPostRnd	0	If set to 1, the FPU will detect tininess after rounding otherwise before. For Meiko compatibility, set to 0, otherwise IEEE-754 recommends 1 (post rounding).
FsMuld	0	If set to 1, will enable the FSMULD instruction. If set to 0, an FSMULD instruction will cause an unimplemented trap (%fsr.ftt = 3).

5 Instruction latency

The instruction latency is measured from the assertion of FpOp to FpBusy low. The latency is somewhat data dependent with the minimum, maximum and average timing shown in the table below.

Instruction	Description	Latency min.	Latency avg.	Latency max.
FADDS, FADD, FSUBS, FSUBD	Add/Sub single/double	5	5.5	9
FMULS, FSMULD	Multiply single	5	7.5	11
FMULD	Multiply double	5	8.5	12
FDIVS	Divide single	11	21	32
FDIVD	Divide double	26	31	63
FSQRTS	Square root single	34	34	35
FSQRD	Square root double	61	61	62
FSTOD	Convert single to double		5	
FDTOS	Convert double to single		5	8

6 Design notes

6.1 Generation of underflow exception (*unf*)

Underflow (tininess) can be detected before or after final rounding by setting the TinyPostRnd generic. This should be taken into account when generating test vectors with software like TestFloat.

6.2 Convert Floating point to Integer Instructions (FDTOI, FSTOI)

These instructions convert the floating-point operand on input 2 into a signed 32-bit integer. The result is always rounded toward zero (the RM input is ignored). On positive overflow, MAX INT is returned (0x7FFFFFFF) while on negative overflow, MIN INT (0x80000000) is returned.

7 Timing diagram

